Danny Rivera
A16906333
3/21/2025

COGS 189 Final Project Paper

Introduction and Motivation

EEG signal processing has become an extremely popular neuroimaging technique for understanding neurological disorders. It is a non-invasive procedure that is relatively computationally inexpensive, and provides high temporal resolution, making it a great option. Many developments in understanding disorders like schizophrenia and Alzheimer's have been achieved with this kind of technology. However, many psychiatric conditions remain relatively unexplored in comparison using EEG data, namely anxiety and its related disorders. My project aims to tackle this problem with a machine learning-related approach, using a large public dataset with EEG data related to various psychiatric disorders. In this paper, I will discuss how I attempt binary machine learning classification to differentiate between healthy controls and patients of two different anxiety disorders – social anxiety disorder, and panic disorder. This is important because it can grow interest in analyzing EEG signals for anxiety patients, and potentially help with diagnosis by observing important differences in brain activity.

Related Work

I looked to see if there had been any research done previously on anxiety patients using EEG recordings, as this would help guide me in my project. Previous studies show that differences between anxiety and healthy patients can be observed with EEG signals, particularly in certain frequency bands. One study observed lower spectral power in the theta, alpha, and beta bands for anxiety patients under both closed-eyes and open-eyes states (Demerdzieva & Jordanova, 2011). Some other studies which also focused on resting-state EEG data observed, inversely, increased power in the beta rhythms for people who scored high in anxiety (Diaz et al., 2019), (Isotani et al., 2001). Overall, there is a pattern in that difference in spectral power measured from EEG recordings across different frequencies can be significant biomarkers for people suffering with anxiety. Because of this, I decided to focus my project on EEG data measuring spectral power and looked for datasets that best suited my interests. This brought me to the dataset that I landed on, which is described below.

I originally discovered the dataset on Kaggle, but I found the original source to be from a paper from Frontiers Psychiatry, titled "*Identification of Major Psychiatric Disorders From Resting-State Electroencephalography Using a Machine Learning Approach*" (Park et al, 2021). Collaborators include multiple researchers at Seoul National University and Ewha Womans University, specializing in departments including their respective institutions' Departments of Psychiatry and Behavioral Science, Statistics, and the Institution of Human Behavioral Health Medicine. This makes the dataset a credible one, despite originally being found on Kaggle. This paper also focuses on using machine learning to identify psychiatric disorders, across various different categories. Based on the paper, I decided to use similar models when creating my project, using different implementations, and observing how my results compared to the ones published.

Dataset and Methods

The dataset includes resting-state EEG data previously recorded between January 2011 to December 2018 at the Seoul Metropolitan Government-Seoul National University (SMG-SNU) in Boramae Medical Center, Seoul, South Korea. The recordings were gathered using Compumedics Neuroscan, with 19 EEG channels recorded based on the international 10-20 system, with a sampling rate of 128 Hz, for over 5 minutes. The data originally includes pre-processed EEG data from people across various different conditions, including healthy patients, but also people from various different psychiatric disorders, for a total of 945 patients. This includes broad categories like schizophrenia, mood disorder, anxiety, as well as specific disorders for each category. For example, the anxiety category includes patients with either panic disorder or social anxiety disorder. Along with the 945 rows, there are initially 1149 columns in the dataset. These columns include some demographic values (like sex and age of the patient), information about the date of the recording and type of condition, and two specific categories of features of EEG data, which are power spectral density (PSD) value and functional connectivity (FC) value across different channels. The PSD columns feature PSD values for different frequency bands among different channels. For example, the "AB.A.delta.FP1" column contains the PSD value among the delta band (around 0.5 to 4 Hz) recorded for the Fp1 electrode. The FC columns contain the functional connectivity between two electrodes based on their measured coherence. For example, the "COH.F.gamma.r.O1.s.O2" column measures the coherence value in the gamma band (around 30 to 100 Hz) between the O1 and O2 electrodes. The dataset was publicly uploaded to OSF as a CSV file by the original creators, and was later uploaded to Kaggle.

For the scope of my project, I will focus only on differentiating healthy control patients from the anxiety disorder patients. This limits my dataset to just 202 observations – 95 control patients, 59 panic disorder patients, and 48 social anxiety patients. This creates large class imbalances among my groups, which will be more relevant as I begin processing for machine learning.

After limiting the dataset to just 202 observations, I began feature selection by removing irrelevant columns, such as the demographic columns, columns with NaN values, and the columns containing coherence values. I then added another column titled "label". The purpose of this was to encode the data based on the category labels given for classifying. Control patients were encoded as 0, panic disorder patients were encoded as 1, and social anxiety patients were encoded as 2. Lastly, I removed the columns indicating the condition name ("main.disorder" and "specific.disorder"). This kept only the columns with PSD values and each observation's conditioned label, now encoded.

Various methods were used from the "sklearn" Python package, for implementing models and evaluating their performance. This is a binary classification task, so algorithms were used to compare performance between controls and either panic disorder or social anxiety patients. I used three distinct, supervised-learning machine models: Logistic Regression, Random Forest, and Support Vector Machine (SVM). Logistic Regression is a simple type of algorithm that works especially well for binary classification. Random Forest is another useful supervised learning algorithm, known for helping reduce the risk of overfitting by averaging over predictions from decision trees. Support Vector Machines are a type of linear classifier that is better suited for high-dimensional data, and picks a hyperplane with maximum margin to optimize variance between groups. For these reasons, I decided to add these algorithms to my analysis, as well as to compare results with the original study.

I created separate functions for each of these models, so that they could be run for each anxiety category (using either the integer 1 or 2 to indicate which condition to use). The columns for PSD values had a logarithmic scale, so I performed a log transformation, which would later help with the metrics for classification. I used an 80:20 ratio for splitting training and testing data, with 20% of the data being reserved for testing, and shuffled at random using a seed for reproducibility. Next, I used MinMaxScaler to normalize and transform the data, which also helped with improving the models. Next, I used one of the models specified and fit it on my training dataset. For logistic regression, I used “balanced” as the “class_weight” parameter to account for the unbalanced classes. For SVM, I used a linear kernel. For both Random Forest and SVM, I used the same random seed number (15) to ensure reproducibility, which was also used across the rest of the notebook. Lastly, the function printed a classification report to report metrics from the models used on the data.

Results

As mentioned earlier, classification requires different methods when using unbalanced classes. Accuracy cannot be used with unbalanced classes as it does not account for proportional differences. Therefore, I observed the F1-scores from the classification reports instead. F1-scores are calculated by taking the harmonic mean of precision and recall metrics, and are better suited for classes that are unbalanced. When looking at the first Logistic Regression model between control and panic disorder patients, the F1-score for control patients was 0.76 and 0.71 for panic disorder patients. For the Logistic Regression model between control and social anxiety disorder patients, the F1-scores reported were 0.6 and 0.57, respectively. This indicates that the logistic regression model was better suited for classifying panic disorder patients, although the numbers could still be improved.

The Random Forest model had slightly lower results for binary classification. In the model between control and panic disorder patients, F1-scores were 0.67 and 0.43, respectively. For control and social anxiety disorder patients, F1-scores were 0.77 and 0.53. As shown, the Random Forest model was much less accurate for classifying anxiety patients, especially for panic disorder patients, being essentially as accurate as random chance.

Lastly, the SVM model showed slightly better results for classifying between controls and anxiety patients. When looking at controls versus panic disorder patients, F1-scores were 0.8

and 0.64. When looking at controls versus social anxiety patients, F1-scores were 0.81 and 0.67. Overall, the SVM model indicated slightly better classification for control patients than the logistic regression model, and similar results to said model for anxiety patients of both conditions.

However, much could be done to improve the findings across all models. For example, hyperparameter tuning or k-fold validation should have been used to ensure reliability and better account for the limitations and small number of the dataset. I wanted to include these in my project, but had repeated trouble implementing this in my code, especially with time constraints, so I had to run models based on what had been done so far. Overall, the results from my analysis had poorer performance than the original study, although this is expected considering my limited analysis and the fact that not all code ran the way I intended it to.

Discussion

The fact that the dataset was pre-processed and required little cleaning/wrangling made it very easy to explore the data and run algorithms, the main focus of my project. However, because of time constraints and limits as a one-person group, the scope of my project was still very limited. Much of the effort in this project was spent trying to write code for feature selection and setting up models to prepare for classification. However, as mentioned earlier, much could have been done to improve the performances of my models, particularly in parameter selection and cross-validation.

Another improvement I could add to my project would be to extend the size and categories of the data used, as well as its models. My project focused on a very small subset of the data – only 202 observations out of the original 945, and with only one major category of psychiatric disorders observed. Additionally, I limited my use of algorithms to only Logistic

Regression, Random Forest, and SVM. These are all supervised machine learning techniques, but it would also be interesting to work with various unsupervised methods to compare performance. After spending some more time improving the models used, I could extend my work to the other categories and specific disorders captured in the data, running a larger analysis to observe which models work best for each disorder. This would be a fascinating project, since other types of models may be better suited for capturing the features of very different psychiatric disorders.

Another improvement could potentially work in a different direction, which is to stick with the anxiety data and focus more on the aspects of the brain regions of the data. There were thousands of entries in the original dataset with information on the functional connectivity between channels recorded, but these were excluded for my analysis. While my project focused on using the PSD values for classification, focusing on FC could help reveal important details missed in the analysis related to specific regions and connections of the brain.

One other improvement could be to go from binary classification to multi-classification between controls, panic disorder patients, and anxiety disorder patients. This could help to observe any relationships between the groups, rather than just comparing one of the disorders with the controls repeatedly. This could be visualized with a confusion matrix, showing where the classifiers struggled to properly distinguish between the conditions.

I learned that machine learning algorithms are valuable tools for analyzing neuroimaging data and potentially detecting anomalies in brain activity, which can lead to diagnosis of severe disorders. However, there are strong limitations to each model and many of them depend heavily on its implementation, including their selected features and validation techniques. With more time, I would love to go back and reassess my models, since these basic implementations cannot accurately diagnose psychiatric conditions like anxiety, especially with this very small sample of data.

Danny Rivera
A16906333
3/21/2025

Works Cited

Park SM, Jeong B, Oh DY, Choi C-H, Jung HY, Lee J-Y, Lee D and Choi J-S (2021) Identification of Major Psychiatric Disorders From Resting-State Electroencephalography Using a Machine Learning Approach. *Front. Psychiatry* 12:707581. doi: 10.3389/fpsyt.2021.707581

Díaz, M. H. et al. EEG Beta band frequency domain evaluation for assessing stress and anxiety in resting, eyes closed, basal conditions. *Procedia Comput. Sci.* 162, 974–981. https://doi.org/10.1016/j.procs.2019.12.075 (2019).

Isotani, T. et al. Source localization of EEG activity during hypnotically induced anxiety and relaxation. *Int. J. Psychophysiol.* 41, 143–153. https://doi.org/10.1016/s0167-8760(00)00197-5 (2001).